AP BIOLOGY TOPIC 1.3

Introduction to Biological Macromolecules

UNIT 1: CHEMISTRY OF LIFE (8–11% EXAM WEIGHT)

SYI-1.B | DEHYDRATION SYNTHESIS | HYDROLYSIS

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 1.3 (INTRODUCTION TO BIOLOGICAL MACROMOLECULES)

CED Core Vocabulary

Monomers — Small organic molecular subunits that serve as building blocks.

Polymerization Math — Linking n monomers into a linear polymer requires n − 1 covalent bonds and releases n − 1 molecules of H_2O .

Repeated Dehydration Reactions — Extending a growing polymer chain adds one monomer at a time; each covalent bond formed releases 1 H_2O .

Chemical Mechanism — A water molecule splits into $H_2O \rightarrow H^+ + OH^-$ across the covalent linkage.

Polymers — Large macromolecules constructed by linking repeating monomers together via strong covalent bonds .

Core Principle — Water removed ($-H_2O$) → new covalent bond formed (Monomers → Polymer).

Chemical Mechanism (Enzyme-Catalyzed) — Remove a hydrogen atom (–H / H^+) from the functional group of one monomer.

Biological Roles of Hydrolysis — Digestion in the GI Tract: Hydrolytic enzymes (amylases, proteases, lipases, nucleases) break ingested food macromolecules into absorbable monomers:...

§2. MONOMERS AND POLYMERS: MODULAR ARCHITECTURE

Covalent Linkages

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Monomers and Polymers ➔ **Monomers** ➔ **Polymers** ➔ **Polymerization Math**

- **Monomers:** Small organic molecular subunits that serve as building blocks.
- **Polymers:** Large macromolecules constructed by linking repeating monomers together via **strong covalent bonds**.
- **Polymerization Math:** Linking **n** monomers into a linear polymer requires **n − 1** covalent bonds and releases **n − 1** molecules of **H_2O** .

MACROMOLECULE CLASS	MONOMER / SUBUNIT	COVALENT LINKAGE NAME
Carbohydrates (Polysaccharides)	Monosaccharides (e.g., Glucose)	**Glycosidic linkage** (**–C–O–C–**)
Proteins (Polypeptides)	Amino acids (20 types)	**Peptide bond** (**–CO–NH–**)
Nucleic Acids (DNA / RNA)	Nucleotides	**Phosphodiester bond** (**5′→3′**)
Lipids *(non-polymeric)*	Glycerol + Fatty acids	**Ester linkage** (**–COO–**)

[EXAMPLE & BIOLOGICAL APPLICATION] **Monosaccharides →** polysaccharides (starch/glycogen); **amino acids →** proteins (polypeptides); **nucleotides →** nucleic acids (DNA/RNA).

[CORE AXIOM / REMEMBER] **Polymer = "beads on a string"** (monomers = individual beads; covalent bonds = the string connecting them).

§4. HYDROLYSIS: CLEAVING POLYMERS WITH WATER

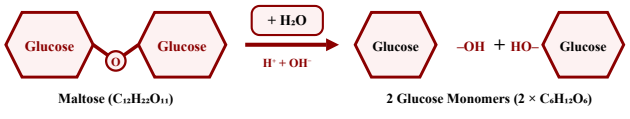
Bond Breaking

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Hydrolysis ➔ **Core Principle** ➔ **Chemical Mechanism** ➔ **Biological Roles of Hydrolysis**

- **Core Principle:** **Water added ($+H_2O$) → covalent bond broken** (Polymer → smaller units / individual monomers). *Hydro* = water, *lysis* = to split.
- **Chemical Mechanism:**
  - A water molecule splits into **$H_2O \rightarrow H^+ + OH^-$** across the covalent linkage.
  - **H^+** (**–H**) attaches to one released monomer.
  - **OH^-** (**–OH**) attaches to the other monomer, restoring both functional groups.
- **Biological Roles of Hydrolysis:**
  - **Digestion in the GI Tract:** Hydrolytic enzymes (amylases, proteases, lipases, nucleases) break ingested food macromolecules into absorbable monomers:
    - **Carbohydrates** → monosaccharides (glucose)
    - **Proteins** → amino acids
    - **Lipids** → fatty acids + glycerol
    - **Nucleic acids** → nucleotides
  - **Intracellular Recycling (Autophagy / Lysosomes):** Breaks worn-out cellular macromolecules back into **reusable monomers** to build new cell structures.

Hydrolysis Visualized: Disaccharide (Maltose) → 2 Glucose Monomers



Maltose ($C_{12}H_{22}O_{11}$) 2 Glucose Monomers (2 × $C_6H_{12}O_6$)

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§3. DEHYDRATION SYNTHESIS (CONDENSATION REACTION)

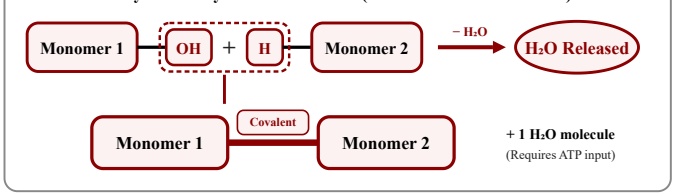
Bond Formation

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Dehydration Synthesis (Condensation Reaction) ➔ **Core Principle** ➔ **Repeated Dehydration Reactions** ➔ **Chemical Mechanism (Enzyme-Catalyzed)**

- **Core Principle:** **Water removed (** $-H_2O$ **) → new covalent bond formed** (Monomers → Polymer).
- **Repeated Dehydration Reactions:** Extending a growing polymer chain adds one monomer at a time; **each covalent bond formed releases 1** H_2O.
- **Chemical Mechanism (Enzyme-Catalyzed):**
  - Remove a hydrogen atom ($-H$ / H^+) from the functional group of one monomer.
  - Remove a hydroxyl group ($-OH$ / OH^-) from the adjacent monomer.
  - $H^+ + OH^- \rightarrow H_2O$ is released as a byproduct, and the two monomers share electrons to form a new **covalent bond**.

Dehydration Synthesis Visualized (Anabolic Bond Formation)



[CORE AXIOM / REMEMBER] **Dehydration = "glue reaction"** (removes water to stick monomers together into a polymer).

§5. ENERGETICS & ATP COUPLING IN POLYMERIZATION

Thermodynamics

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Energetics & ATP Coupling in Polymerization ➔ **Hydrolysis = Exergonic (ΔG < 0)** ➔ **Energy Coupling via ATP Hydrolysis** ➔ **powers endergonic dehydration synthesis**

- **Dehydration Synthesis = Endergonic (** $\Delta G > 0$ **):**
  - Anabolic assembly increases molecular order (decreases entropy, $-\Delta S$) and **requires free energy input** (usually coupled to **ATP hydrolysis**).
- **Hydrolysis = Exergonic (** $\Delta G < 0$ **):**
  - Catabolic cleavage breaks polymers into smaller units, increasing entropy ($+\Delta S$) and **releasing free energy**.
- **Energy Coupling via ATP Hydrolysis:**
  - **ATP itself is broken down by hydrolysis:**

$$ATP + H_2O \rightarrow ADP + P_i + \text{Free Energy } (\Delta G = -30.5 \text{ kJ/mol})$$

- That released free energy from ATP hydrolysis **powers endergonic dehydration synthesis** reactions (e.g., protein translation at ribosomes, DNA replication, glycogen synthesis).

[MECHANISTIC NOTE] **Yes, it's counterintuitive at the macromolecular level:** Building biological polymers via dehydration synthesis **costs energy (endergonic)**, whereas hydrolyzing macromolecules into monomers **releases energy (exergonic)**!

[AP EXAM FRQ SCORING STRATEGY] **Stoichiometry Calculation Trap:** If an AP question asks for the chemical formula of a trisaccharide made of 3 hexose sugars ($C_6H_{12}O_6$), do NOT simply multiply by 3 ($C_{18}H_{36}O_{18}$)! Subtract **2** H_2O ($n - 1 = 2$ bonds formed) → correct formula is $C_{18}H_{32}O_{16}$.

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 1.3 [MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]

Topic 1.3: Introduction to Biological Macromolecules ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Lab Context: Macromolecule Indicator Assays (Benedict's for reducing sugars, Lugol's Iodine for starch, Biuret for peptide bonds) & Buffer Titration Curves. Independent Variable (IV): pH or temperature treatment; Dependent Variable (DV): absorbance/colorimetric change or structural stability; Negative Control: deionized H_2O. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Stoichiometric monomer-to-polymer condensation: for n monomers joined in a linear chain, (n − 1) covalent bonds form and (n − 1) H_2O molecules are released. pH scale is logarithmic: $pH = -\log_{10}[H^+]$; each 1-unit decrease in pH represents a 10-fold increase in $[H^+]$. Caloric density: Lipids (~9 kcal/g) vs. Carbohydrates/Proteins (~4 kcal/g). **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Molecular Disruption Case Study: Missense mutation in the β-globin gene (Glu6Val in Sickle-Cell Disease) replaces a polar/charged glutamic acid R-group with a nonpolar hydrophobic valine on the protein exterior, driving abnormal quaternary hydrophobic aggregation into rigid fibers under deoxygenated conditions. Similarly, extreme fever (>40°C) or acidosis (pH < 7.0) disrupts weak hydrogen and ionic bonds in tertiary (3°) protein structure while leaving primary (1°) peptide bonds intact.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 1.3 to prove why the claim holds.

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